| NOT MEASUREMENT |
| SENSITIVE |

MIL-R-28567C
8 March 1989
SUPERSEDING
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24 December 1981

MILITARY SPECIFICATION

ROLLER, MOTORIZED, VIBRATING DRUM, ARTICULATED, PNEUMATIC TIRE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a motorized roller, equipped with a vibrating steel drum and two pneumatic tires, mounted on an articulated frame, and powered by a diesel engine through a hydrostatic transmission.

1.2 Classification. The roller will be of the following types, as specified (see 6.2):

Type I - Type I roller has a tamping foot drum, optional smooth drum or one of each drum, interchangeable, (when specified, see 6.2), and non-directional traction tires, for compacting soil, subbase, gravel, and other non-asphaltic surfaces.
Type II - Type II roller has a smooth drum, sprinkler system, and smooth tread tires for compacting asphaltic pavements.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-B-131 - Battery Storage (Vehicular, Ignition, Lighting, and Starting).

MILITARY

MIL-R-3075 - Rollers, Motorized, Road, Diesel- or Gasoline-Engine-Driven, Packaging of.
MIL-H-13531 - Hose, Rubber, and Hose Assembly, Rubber (Hydraulic, Flexible).

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the document cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1557 - Moisture-Density Relations of Soils and Soil-Aggregate Mixtures using 10-lb Rammer and 18-in Drop, Test Methods for.
ASTM D 2487 - Classification of Soils for Engineering Purposes, Test Method for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J88 - Sound Measurement, Earthmoving Machinery, Exterior.
SAE J534 - Fittings, Lubrication.
SAE J919 - Sound Measurement, Earthmoving Machinery, Operator, Singular Type.
SAE J1136 - Braking Performance - Rollers and Compactors.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The roller shall include a frame, an engine and power transmission assembly, a drum assembly, pneumatic tires, an operator's station and controls, accessories, and auxiliary systems.

3.2 First article roller. When specified (see 6.2), the contractor shall furnish one roller for first article inspection (see 4.2.1 and 6.4).

3.1 Standard commercial product. The roller shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the roller being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

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3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.5 Interchangeability. All rollers of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Maintainability. The roller shall be designed so that components are readily accessible for repair or replacement, with minimum removal or disturbance of adjacent parts or components, using general-purpose tools. Operating parts, accessories, and drain outlets shall be readily accessible for regular maintenance service.

3.7 Safety. All reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of equipment, including exhaust pipes and other lines which may be subject to high temperatures, exposed to contact by personnel, or which create a hazard. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Exhaust or discharges from equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. All points requiring lubrication during operation shall have fittings located or guarded so as to be accessible without hazardous exposure to personnel performing these operations. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. The noise level in the area occupied by the operator shall not exceed 85 decibels (dB) when tested in accordance with SAE J919. Exterior noise shall not exceed 88 dB, verified in accordance with SAE J88.

3.8 Performance. The type II roller shall be capable of compacting hot and cold mixed asphaltic concrete having a loose depth thickness of not less than 3 inches to a smooth compacted surface in not more than five passes. The roller shall be required to compact fine and coarse grained soils as classified in accordance with ASTM D 2487 to a density of 90 percent modified proctor as determined in accordance with ASTM D 1557. The roller shall withstand the strain imposed upon it without any damage or deformation when operating at maximum speeds and as specified herein.

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3.9 Weight and dimensions. Equipped with either drum and without ballast, the roller shall weigh not less than 20,000 pounds (lb). The roller shall conform to the dimensions as specified in table I.

TABLE I. Roller dimensions.

ITEM	DIMENSIONS	
	MINIMUM	MAXIMUM
Overall dimensions:		
length (inches (in))	198	260
width (in)	90	106
Drum dimensions:		
length (in)	80	90
diameter (in)	54	-
head thickness (in)	3/4	-
Wheel base (in)	102	130
Outside turning radius (in)	-	360
Curb clearance (in)	15	-

3.10 Frame and auxiliary systems. The frame shall be articulated to facilitate steering. The hinge point shall be located at or near the center of the roller and on the longitudinal axis of the frame. The roller design shall allow the vibrating drum to oscillate not less than 10 degrees above and below the drum's longitudinal axis. The frame shall be mounted on two drive wheels on the rear and a single vibrating drum on the front. The manufacturer shall provide a lockout device to preclude the frame from articulating during servicing or when loaded on a carrier for transport.

3.10.1 Rims, tires, and tubes. Tire and rim sizes shall conform to the Tire and Rim Association's recommendations for the type and size of the tires furnished. Tires for type I roller shall be of the wide base type with hard rock lug treads. Tires for type II roller shall be of the smooth type. Tires shall have a rated capacity at least equal to the load measured at each wheel at the ground. If tube type tires are furnished, tubes shall be of the heavy-duty type and of mating size with the tires. Tires and tubes shall be equipped with air-water valves.

3.10.2 Brakes. The service braking system, secondary stopping system, and the parking system shall conform to the performance requirements of SAE J1136.

3.10.3 Lighting system. A complete 12-volt (V) lighting system shall be provided. The lighting system shall include two forward- and two rearward-directed sealed beam floodlights. All wiring shall be protected against wearing and weathering and shall be so secured that no section exceeding 3 feet in length is unsupported. Grommets shall be used where wiring passes through structural steel sections and sheet metal to prevent chaffing of the cable.

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3.10.4 Hydraulic system. The hydraulic system shall be complete with all operating accessories, including pumps, reservoir, hydraulic cylinders, valves, pressure relief valves, filters, hoses, piping, and fittings. A wiper shall be provided on all cylinders adjacent to the piston rod seal to prevent dirt and abrasives from damaging the seal or entering the cylinders. The hydraulic system shall be equipped with a full flow, 10 micron oil filter accessible for cleaning. Vents shall be equipped with filter elements to prevent dirt from entering the reservoir with the air. The suction pickup shall be above the reservoir bottom. The return line shall terminate below the minimum oil level of the reservoir. The drain plug shall be located at the lowest point in the tank and shall be of the magnetic type. All high pressure hydraulic hoses and fittings capable of withstanding a bursting test pressure of not less than four times the working pressure, and a maximum operating pressure no less than two times the working pressure. High pressure hydraulic hoses shall have physical qualities equal to or conforming to MIL-H-13531, type I or type II, as applicable. Hydraulic oil shall be of the noncorrosive type, with the characteristics to withstand a temperature range from 10 degrees Fahrenheit (^{o}F) to 325°F.

3.11 Power transmission system. The roller shall be diesel-engine driven, with a hydrostatic transmission matched with the engine for power and speed.

3.11.1 Engine. The roller shall be powered by a rear-mounted diesel engine having horsepower, torque, and speed characteristics to meet all roller performance requirements specified herein. The diesel engine shall operate in any ambient temperatures from -20°F to +120°F. The engine shall be ready for full load operation in less than 15 minutes. When an ether priming system is furnished, it shall be of the measured shot type with storage capacity of not less than 12 fluid ounces. The engine shall have an exhaust system with muffler. The engine shall be furnished complete with at least the following accessories:

a. Fuel tank of sufficient capacity for 8 hours normal operation.
b. Two-stage dry type intake air cleaner with service indicator.
c. Battery mounting location painted with acid-resistant paint.
d. A 12-V negative ground direct current cranking system with battery in accordance with W-B-131. The battery shall have a capacity of not less than 100 ampere hours at a 20-hour rate.
e. Battery charging alternator with a rating of not less than 35 amperes.
f. Indicating control devices of the visual or audible type for high coolant temperature and low lubricating oil pressure.
g. Engine housing.
h. Automatic switch or device to preclude accidental starts when transmission or drive system is in any mode except neutral.
i. Instrument and control panel located at the operator's station and equipped with, but not limited to, the following indicators and controls:

    (1) Coolant temperature indicator.
    (2) Liquid fuel quantity indicator.
    (3) Lubricating oil pressure indicator.
    (4) Tachometer.

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(5) Manual speed control.
(6) Emergency shutdown device, when two-cycle engine is furnished.
(7) Ignition or start switch.
(8) Hour meter, totalizing mechanism of not less than 9999 hours.
(9) Normal shutdown device.
(10) Battery charging indicator.
(11) Air cleaner indicator.

3.11.2 Transmission. Unless otherwise specified (see 6.2), power from the engine shall be transmitted to the drive wheels and the front drum by a hydrostatic drive, a selective gear transmission having not less than two speeds forward and one reverse speed, and a nonslip differential. The roller shall operate at any speed from 0.5 miles per hour (mph) to not less than 6 mph while traveling on a level paved surface. The roller shall be able to negotiate a 15 percent slope.

3.12 Drum. As specified (see 6.2), type I rollers shall be equipped with a tamping foot drum conforming to 3.12.2, smooth drum conforming to 3.12.1, or one of each style with interchangeability design. Type II rollers shall be equipped with a smooth drum conforming to 3.12.1.

3.12.1 Smooth drum. The length and diameter of the drum shall be as specified in table I. A drum equipped with an internal vibrating mechanism shall have an outside diameter of not less than 54 inches.

3.12.2 Tamping foot drum. The tamping foot drum shall have length and diameter as specified in table I. The tamping foot drum shall be furnished with an independent axle assembly mounted on antifriction type bearings. The quantity and type of tamping feet provided on the drum shall be the manufacturer's standard as designed for use on the materials described (see 1.2, type I).

3.12.3 Drum cleaner. Replaceable and adjustable drum cleaners shall be provided for the type I rollers. The cleaners shall extend over the full length of the drum for smooth drums and between feet for the tamping foot drums. Drum cleaners shall prevent accumulation of material on the drum when the roller is propelled in either the forward or reverse direction.

3.12.4 Vibrating mechanism. A variable frequency vibrating mechanism shall be provided. The vibrating mechanism shall have a minimum frequency of not less than 1600 $\pm$ 200 vibrations per minute (vpm) for type I and 2100 $\pm$ 200 vpm for type II rollers including a minimum of two frequency selections. When operating at the manufacturer's recommended maximum continuous frequency, the centrifugal force (cf) generated by the vibrating mechanism shall be not less than 23,000 lb for type I rollers. Type II rollers shall be capable of generating not less than 35,000 lb cf at operator's selection. The following formula shall be used to compute the minimum cf requirement specified herein:

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$$CF = 0.0000284\ wrn^2$$

where,

CF = centrifugal force
w = weight of eccentric revolving body in lb
r = mean radius in inches
n = number of revolutions per minute (rpm)

All rotating shafts shall be mounted on heavy-duty bronze or antifriction type bearings. The vibrating mechanism shall be driven by the roller's engine through a hydrostatic drive. Controls for the engagement, disengagement, and frequency of the vibrating mechanism shall be provided in the operator's station.

3.12.5 Sprinkler system. Type II rollers shall be equipped with a sprinkler system for wetting the drum and rear tires. The sprinkler system shall include all piping, fittings, controls, and water tanks necessary to wet the full width of the drum and full width of the tires. Piping and fittings shall be nonferrous type, with the characteristics necessary to withstand the vibration transmitted from the roller under all operating conditions. Separate controls shall be provided for the drum and the rear tires. The tanks shall have a total capacity of not less than 150 gallons. The tanks shall be equipped with vents and drain plugs that shall allow complete drainage of water.

3.13 Operator's station. The roller shall be provided with an operator's station. The station shall be furnished with all controls necessary to operate the roller. The station shall be equipped with a durable operator's seat with a waterproof covering. The station shall include those instruments specified in 3.11.1i and a frequency meter indicating vpm of the vibrating mechanism. All instruments shall be visible from the normal operating position. All controls shall be within reach of the operator when in the normal operating position. Steps or ladder rungs and handholds shall be furnished as necessary to provide safe and easy access for the operator to the station. The station deck and steps shall have an antiskid surface. The station shall be isolated from the vibrations generated by the vibrating mechanism.

3.14 Toolbox. The roller shall be equipped with a metal toolbox of sufficient size to accommodate the operating and maintenance tools normally furnished with the roller.

3.15 Lifting and tiedown attachments. When specified (see 6.2), the roller shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. Lifting attachments shall be designed to preclude the use of spreader bars. A nonferrous transportation plate shall be provided and mechanically attached to the roller. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the roller on the carrier when shipped.

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3.16 Instruction plates. The roller shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.17 Identification plate. An identification plate will be furnished by the contracting officer for each roller. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each roller in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.18 Servicing and adjusting. Prior to acceptance of the roller by the Government, the contractor shall service and adjust the roller for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.
c. Proper functioning of all lighting and electrical systems.
d. Wheel alinement, when required.
e. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
f. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The roller shall be conspicuously tagged to identify the lubricants and their temperature range.

3.19 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2), in standard yellow nearest match to No. 13538 conforming to FED-STD-595. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

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3.20 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 pound-force per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.21 Workmanship.

3.21.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.21.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.21.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the diameter of rivet. Rivet heads shall be ful neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.21.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.22 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one roller when a first article is required (see 3.2, 6.2, and 6.4). The first article produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. This inspection shall include the examination of 4.3 and the tests of 4.4. The purpose of the inspection shall be to determine conformity with the requirements of the contract. Acceptance of the first article shall not constitute a waiver by the Government of its right under the provisions of the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.6.

4.3 Examination. Each roller shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Performance tests. The first article shall receive the tests of 4.4.1, 4.4.2, and 4.4.3. Each production unit shall receive the test of 4.4.2. Failure to pass any test shall constitute cause for rejection.

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4.4.1 Maneuverability. The first article shall be tested for maneuverability on a test course. The test course shall have a relatively level grade with a rough surface. Degree of roughness shall be sufficient to demonstrate the drum's ability to oscillate to its maximum limit. Course length shall be not less than 200 feet. Width shall be sufficient to allow roller to turn around without reversing the traction drive. The operator shall maintain the highest possible travel speed as dictated by the roughness of the course and without jeopardizing good safety practices. The vibrating mechanism shall operate at a frequency of not less than that specified for the applicable type of roller being tested, in accordance with 3.12.4. The roller shall perform the following maneuvers for not less than 20 minutes each:

a. Figure-eight turns using the shortest possible turning radius.
b. Travel in forward gear.
c. Travel in reverse gear.

Throughout the test, the roller shall be observed to verify compliance with the requirements as specified in 3.9, 3.10, 3.11.2, and 3.12.4.

4.4.2 Brake system test. The brake system shall be tested to determine functionability and conformance to the requirements specified in 3.10.2.

4.4.3 Sprinkler test. Type II rollers shall be tested to determine conformance to the requirements specified in 3.12.5. The water tank(s) shall be filled to capacity. The roller shall then be driven on a bituminous or asphalt paved surface throughout the test. The sprinkler shall be operated for not less than 30 minutes while the roller is traveling at not less than 5 mph. All controls shall be operated as many times as necessary to determine ease of operation, effectiveness, and responsiveness. Upon completion of the test, remaining water shall be drained from the tank(s).

4.4.4 Lifting and tying down attachments test. When required (see 6.2), the roller shall be tested to verify that the attachments conform to the requirements specified in 3.15.

4.4.5 Production sample. Upon acceptance of the first article, the first article shall remain at the manufacturing facility as a production sample, and shall be the last roller delivered on the contract. The first article shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first article in a serviceable condition for the duration of the contract.

4.5 Production operational tests. Each production roller shall be operated for one hour at speeds between 1 mph and top speed at applicable vibrating frequency for the type of roller being tested. All controls shall be operated as many times as necessary to determine ease of operation, effectiveness, responsiveness, and that the mechanisms actuated by the controls operate promptly, fully and without restriction, malfunction, or excessive vibration.

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The hydraulic system, including hoses, fittings, and seals shall be examined for evidence of leakage or failure. Gear cases, bearings, shafts, and other moving parts shall be factually examined for excessive heating or abnormal operation insofar as is practicable. Failure to pass any phase of this test shall be cause for rejection.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-R-3075. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-R-3075 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The type I roller is primarily intended for compacting aggregate and earth. The type II roller is primarily intended for compacting hot and cold layed bituminous materials.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type of roller (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When one of each style roller is required with interchangeable design (see 1.2 and 3.12).
e. When type I roller is to be furnished with smooth drum (see 1.2 and 3.12.1).
f. When tamping foot drum is required for type I roller (see 1.2 and 3.12.2).
g. When first article is to be furnished (see 3.2, 4.2.1, and 6.4).
h. If transmission is to be other than specified (3.11.2).
i. When lifting and tiedown attachments are required (see 3.15 and 4.4.4).
j. Finish color of roller (see 3.19).
k. When a preproduction pack is to be furnished for inspection and time frame required for submission (see 4.6).
l. Level of preservation and level of packing required (see 5.1).

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6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the roller should be tested and should be a first production roller or a standard roller from the contractor's current inventory. The contracting officer should include specific instructions in the contract regarding arrangements for examination, test, and approval of the first article (see 3.2, 4.2.1 and 6.2).

6.5 Subject term (key word) listing.

Compactor, earth
Compactor, bituminous pavement

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - YD

Review activity:
DLA - CS

Preparing activity:
Navy - YD

(Project 3895-N243)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-28567C	ROLLER, MOTORIZED, VIBRATING DRUM, ARTICULATED, PNEUMATIC TI

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

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